SemBind: Inspectable Self-Supervised and Spectral Placement for Reference-Based Speech Watermarking

Sanjeev Kumar, Biswajit Maity, Banashree Mandal, and Aditya Kumar Sahu

***Abstract*—Speech watermarking schemes decide where to embed from signal energy or leave the decision implicit inside a trained network, so the effect of placement on speech content is rarely measured. This letter is a controlled study of that effect, built on SemBind, a reference-based STFT watermark whose placement is explicit: a frozen HuBERT model gives a closed-form leave-one-out contribution map from one forward pass, which is fused with spectral suitability to place a 256-bit, three-replica, SHA-256-bound payload. Holding the embedding rule fixed, we compare placement rules across a strength sweep on LibriSpeech. On the full 2,620-utterance LibriSpeech test-clean corpus, a six-level sweep from 87 to 20 dB evaluates five placement rules, including random placement. At target 40 dB, joint and spectral-only placement change 8.85% and 8.28% of Whisper tiny.en transcripts, compared with 18.36% for high-contribution and 16.98% for random placement. At 30 dB, the corresponding rates are 21.18%, 19.12%, 30.46%, and 29.69%. Low-contribution placement changes significantly fewer transcripts than high-contribution placement at 40–27 dB and than random placement at 40–20 dB, so the HuBERT map is informative; however, joint and spectral-only placement are statistically indistinguishable at every strength, so local energy, not representational redundancy, is the stronger placement criterion.**

***Index Terms*—Speech watermarking, embedding-location selection, self-supervised learning, HuBERT, content preservation, provenance.**

I. Introduction

WATERMARKING supports ownership identification and provenance of speech recordings [1]. Besides the payload and the extractor, a scheme must decide *where* to embed. Classical spread-spectrum [2], [3], echo, and patchwork schemes choose locations from energy, transform statistics, masking, or fixed bands [4]–[7]. Neural schemes [8]–[13] learn the embedder and detector jointly, and recent letters continue with adversarial, generative, and reinforcement-learned designs [14]–[16]. In all of these, placement cannot be varied independently of encoding, so its effect on content is rarely isolated.

Self-supervised models such as HuBERT [17] and wav2vec 2.0 [18] encode phonetic and speaker content per frame, which suggests a content-aware alternative to energy-based placement: embed where a frame contributes least to the utterance representation. This letter tests that hypothesis directly; to our knowledge, it is the first study of speech watermarking that varies placement alone while holding the payload, embedding rule, and extractor fixed.

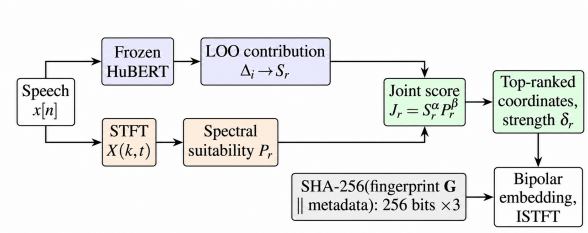


Fig. 1. SemBind embedding. HuBERT only produces the contribution map and the fingerprint; embedding is in the STFT domain.

The contributions are: (i) SemBind, a reference-based watermark with an explicit placement stage, built on a closed-form leave-one-out contribution map from one HuBERT pass; (ii) a controlled comparison in which only placement varies, from 87 to 20 dB SNR, with classical and neural baselines; and (iii) the finding, explained by a local-SNR analysis, that self-supervised contribution is informative but weaker than local energy, so representational redundancy alone does not identify the safest locations. Blind extraction and robustness are out of scope.

II. Proposed Method

A. Representations

Let $X(k,t)$ be the STFT of the 16-kHz input $x[n] \in [-1,1]$ (periodic Hann window, $N = 1024$, hop $H = 320$, centered reflective padding, no normalization, 513 one-sided bins). The frozen HuBERT-base model [17] (`hubert-base-ls960`, last layer) gives frame representations $\mathbf{E}_1, \ldots, \mathbf{E}_T \in \mathbb{R}^{768}$ with mean $\mathbf{G}$. Removing frame i gives, in closed form, the contribution Δ_i and representation suitability S_i:

$$\Delta_i = 1 - \cos\Big(\mathbf{G}, \frac{T\mathbf{G} - \mathbf{E}_i}{T-1}\Big), \qquad S_i = 1 - \mathrm{minmax}(\Delta_i). \tag{1}$$

This needs one forward pass and $O(Td)$ operations. A frame with small Δ_i (e.g., a pause) is redundant for the utterance summary and is expected to be safer to perturb; high-contribution placement is the negative control.

(Corresponding author: Biswajit Maity.)

S. Kumar, B. Maity, and A. K. Sahu are with the Department of Computer Science and Engineering, SRM University-AP, Amaravati 522 240, Andhra Pradesh, India (e-mail: sanjeevkumar_s@srmap.edu.in; biswajit.m@srmap.edu.in; aditya.s@srmap.edu.in).

B. Mandal is with the Department of CSE (AIML), CSBS, IEM, Salt Lake, Kolkata, India (e-mail: bana2k11.mandal@gmail.com).

B. Spectral Suitability and Joint Placement

HuBERT frame r (receptive field centered at sample $rH + 200$) maps to the single centered STFT frame $t = r + 1$, since both use a 20-ms stride. With admissible bins $\mathcal{F} = \{1, \dots, 511\}$ (DC and Nyquist excluded),

$$P_r = \operatorname{minmax}\Big(\tfrac{1}{|\mathcal{F}|}\textstyle\sum_{k\in\mathcal{F}} \log\big(1 + |X(k, r{+}1)|\big)\Big) \qquad (2)$$

measures local signal headroom (not a psychoacoustic model). Placement and strength follow from the joint score

$$J_r = S_r^{\alpha} P_r^{\beta}, \qquad \delta_r = \delta_{\min} + (\delta_{\max} - \delta_{\min}) J_r, \qquad (3)$$

where $\alpha = 0.7$ and $\beta = 0.3$ weight the two criteria and $\delta_{\min} = 0.002$, $\delta_{\max} = 0.008$ (unnormalized STFT units). Spectral-only, representation-only, and high-contribution placement replace J_r by P_r, S_r, and $1 - S_r$.

Allocation with replica separation. The top $\lceil \rho T \rceil$ regions by J_r ($\rho = 0.5$ is the selected fraction) each contribute $m = \lceil 256R / \lceil \rho T \rceil \rceil$ distinct bins drawn from $\mathcal{F}$ by a keyed pseudo-random generator, appended in ranking order to a list $\mathcal{C}$ of $256R$ coordinates ($R = 3$). Replica q of bit i takes entry $i + 256q$. Because each region occupies m consecutive entries and replicas are 256 entries apart, the three replicas of a bit lie in distinct regions whenever $m \le 256$, i.e., $\lceil \rho T \rceil \ge 3$, which holds for any utterance longer than about 0.12 s.

C. Payload, Embedding, and Verification

The payload $\mathbf{b} \in \{0,1\}^{256}$ is the SHA-256 digest [19] of a canonical JSON object holding $\mathbf{G}$ (four decimals), an owner identifier o, and a UTC timestamp τ. Each coordinate $c \in \mathcal{C}$ carrying bit b_i in region r is modified, and each replica is read back against the reference, as

$$\operatorname{Re} X'(c) = \operatorname{Re} X(c) + (2b_i - 1)\,\delta_r, \qquad (4)$$

$$\hat{b}_{i,q} = \mathbf{1}\big[\operatorname{Re}\widetilde{X}(c_{i,q}) > \operatorname{Re} X(c_{i,q})\big], \qquad (5)$$

so bit 1 raises and bit 0 lowers the coefficient. Verification (Algorithm 2) needs the clean reference x and the trusted metadata (o, τ, κ) from the owner's provenance record: it recomputes $\mathcal{C}$ and the expected $\mathbf{b}$ from x, so no coordinates are stored. SHA-256 binds content and metadata for integrity; it does not by itself prove who embedded the mark, which would require a keyed MAC or a signature. The default configuration uses the adaptive δ_r of (3); the controlled experiments fix $\delta_r \equiv \delta$ to isolate placement. Selection and embedding take 52 ms per utterance on one CPU core beyond one HuBERT pass (real-time factor 0.007).

Algorithm 1 SemBind embedding

Require: clean speech x; owner o, timestamp τ, key κ
Ensure: watermarked speech x_w
$X \leftarrow \mathrm{STFT}(x)$; $X' \leftarrow X$; $\mathbf{E}_{1:T} \leftarrow \mathrm{HuBERT}(x)$
$S_r, P_r, J_r, \delta_r \leftarrow$ (1)–(3)
$\mathbf{b} \leftarrow \mathrm{SHA256}(\mathbf{G} \,\|\, o \,\|\, \tau)$
$\mathcal{C} \leftarrow \textsc{Allocate}(J, \kappa)$ ▷ m bins per top region
for $i = 0, \dots, 255$ **and** $q = 0, \dots, R-1$ **do**
  update $X'(c_{i,q})$, $c_{i,q} = \mathcal{C}_{i+256q}$, by (4)
end for
return $x_w \leftarrow \mathrm{ISTFT}(X')$

Algorithm 2 SemBind reference-based verification

Require: observed $\tilde{x}$; clean reference x; trusted (o, τ, κ)
Ensure: ACCEPT or REJECT
recompute $\mathbf{G}$, J_r, and $\mathcal{C} \leftarrow \textsc{Allocate}(J, \kappa)$ from x
$\mathbf{b} \leftarrow \mathrm{SHA256}(\mathbf{G} \,\|\, o \,\|\, \tau)$ ▷ expected payload
$\widetilde{X} \leftarrow \mathrm{STFT}(\tilde{x})$; $X \leftarrow \mathrm{STFT}(x)$
for $i = 0, \dots, 255$ **do**
  $\hat{b}_i \leftarrow \mathrm{majority}_q\, \hat{b}_{i,q}$ by (5)
end for
return ACCEPT if $\hat{\mathbf{b}} = \mathbf{b}$, else REJECT

III. Experimental Setup

LibriSpeech test-clean [20] contains 2,620 utterances from 40 speakers. The primary placement sweep evaluates all 2,620 utterances under five deterministic rules: joint (J_r), spectral-only (P_r), low contribution (S_r), high contribution ($1 - S_r$, a negative control), and random placement. For each rule, the embedding strength is fixed within a condition and is swept over target SNRs of 87, 61, 40, 30, 27, and 20 dB. Content damage is measured by the number and percentage of utterances whose Whisper tiny.en [21] transcript differs from the clean reference. Decoding is greedy, with no temperature fallback and no previous-text conditioning. Rules are compared with two-sided two-proportion z-tests on these counts; because every rule is applied to the same utterances, this unpaired test is conservative. The full-corpus strength sweep measures content damage and waveform SNR; it does not re-evaluate payload recovery at every strength. The fixed-strength recovery and baseline comparisons use the deterministic 200-utterance subset. Baselines are a reference-based DWT spread-spectrum scheme after Cox et al. [4] with the same payload, three replicas, and SNR, and AudioSeal [12] at its native operating point.

IV. Results

A. Recovery and Fidelity

At 87 dB every STFT rule recovers the full payload in every utterance, whereas DWT-SS with the same payload, replicas, and SNR fails on 14 of 200, so in this matched setting coefficient-level bipolar embedding is the more reliable carrier. PESQ and STOI are at their ceilings, and at the 87-dB target at most 10 of 2,620 transcripts (0.38%) change under any rule (Fig. 2), so at transparent strength placement has no measurable content cost.

Table II positions SemBind among the baselines rather than ranking them. AudioSeal provides blind, processing-robust detection of a 16-bit message at 27 dB; SemBind verifies a 256-bit payload at 87 dB only when the reference is available. The SNR gap reflects these different goals, not overall superiority; SemBind's distinguishing property is that its placement can be read and varied directly.

B. When Does Placement Matter?

Across the full corpus, transcript changes are low but nonzero at the 87 dB target (7–10 of 2,620 utterances, depending on placement) and remain below 1.4% at 61 dB. Placement

TABLE I
RECOVERY AND FIDELITY ON 200 UTTERANCES. STFT ROWS USE IDENTICAL FIXED STRENGTH ($\delta = 0.005$).

Method	Recovery	SNR (dB)↑	PESQ↑	STOI↑
SemBind (joint)	100%	86.99	4.64	1.0000
Spectral-only	100%	87.00	4.64	1.0000
Low contribution	100%	87.14	4.64	1.0000
High contribution	100%	87.05	4.64	1.0000
Random	100%	87.07	4.64	1.0000
DWT-SS [4]	93%	86.99	4.64	0.9999
AudioSeal[a] [12]	100%	27.13	4.50	0.9980

[a] Blind detector, 16-bit message (accuracy 100%), native strength.

TABLE II
OPERATING POINTS OF THE COMPARED SCHEMES (200 UTTERANCES). PAYLOAD, EXTRACTION MODE, AND ROBUSTNESS DIFFER, SO THE ROWS ARE NOT A RANKING.

Scheme	Placement	Extraction	Payload	SNR (dB)	Recovery
DWT-SS [4]	fixed band	reference	256 b	86.99	93%
AudioSeal [12]	learned	blind	16 b	27.13	100%
SemBind	explicit	reference	256 b	86.99	100%

effects become substantial from 40 dB downward. At 40 dB, joint and spectral-only placement change 8.85% and 8.28% of transcripts, respectively, compared with 18.36% for high-contribution and 16.98% for random placement. At 30 dB, the corresponding rates are 21.18%, 19.12%, 30.46%, and 29.69%.

Joint placement changes far fewer transcripts than random and high-contribution placement at 40–20 dB ($p < 10^{-7}$ at every strength), but it is statistically indistinguishable from spectral-only placement at every strength ($p \geq 0.06$), although spectral-only is numerically lower at five of six. The present multiplicative fusion therefore matches, but does not improve on, spectral-only placement (Fig. 2).

The HuBERT contribution map is informative on its own. Low-contribution placement changes fewer transcripts than high-contribution placement at every target from 61 to 20 dB (28 vs. 33, 368 vs. 481, 698 vs. 798, 827 vs. 913, 1166 vs. 1226), significantly so at 40, 30, and 27 dB ($p = 2 \times 10^{-5}$, 0.002, 0.012), and fewer than random placement at 40–20 dB ($p \leq 0.046$). It therefore separates safer from riskier frames, but less sharply than spectral suitability.

C. Why Spectral Placement Wins

The explicit placement stage explains the sweep. Low-contribution placement fills the leading and trailing pauses (Fig. 3, 0–0.5 s and after 7.4 s), as the redundancy argument predicts: pauses contribute least to $\mathbf{G}$, so S_r peaks there. But for m_r coordinates of strength δ in frame $t = r+1$, the local signal-to-watermark ratio is

$$\mathrm{SNR}_r = 10 \log_{10} \frac{\sum_{k \in \mathcal{F}} |X(k,t)|^2}{m_r \, \delta^2}, \tag{6}$$

which falls decibel for decibel with frame energy at fixed δ. This first-order expression assumes a fixed δ and only the frame's own energy over $\mathcal{F}$; it ignores overlap-add spreading

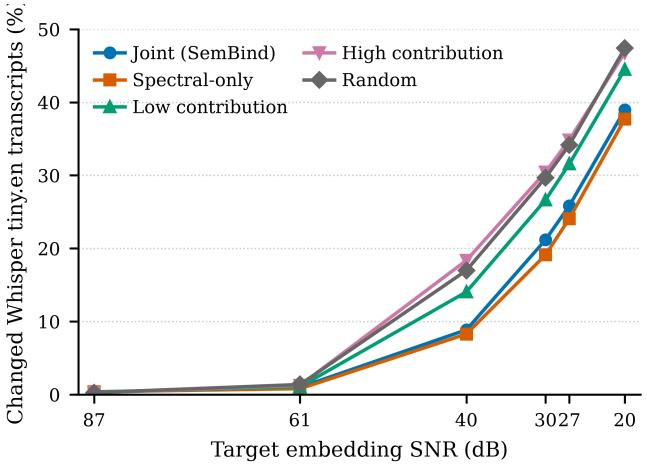


Fig. 2. Full-corpus strength sweep (2,620 utterances per rule, Whisper tiny.en): changed transcripts for the five placement rules. Colors match Fig. 3.

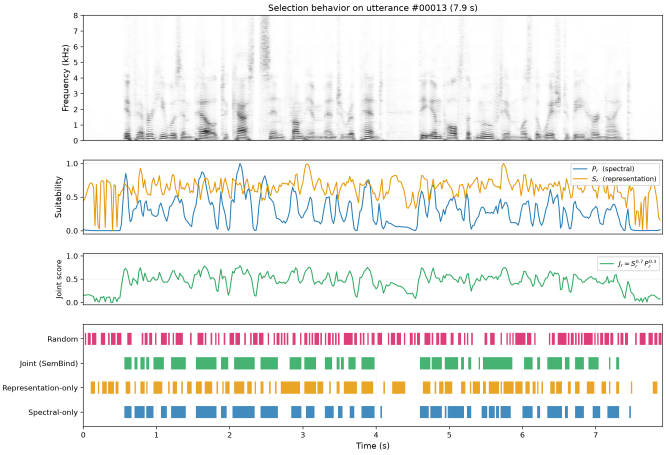


Fig. 3. Placement on one utterance (7.9 s): spectrogram; P_r, S_r, and J_r; and the frames used by each rule. Low-contribution (representation-only) selection fills the pauses; joint and spectral-only selection avoid them because $P_r \approx 0$ there.

into adjacent frames and auditory masking, so it ranks frames rather than predicting ASR errors. A pause therefore receives the most exposed perturbation in the utterance, consistent with the higher counts of low-contribution placement. Because P_r in (2) grows with log-magnitude, ranking by P_r approximately ranks frames by SNR_r: spectral placement is, in effect, local-SNR maximization. It excludes pauses ($P_r \approx 0$), and because $J_r = S_r^{\alpha} P_r^{\beta}$ is multiplicative, joint placement excludes them too and, as Fig. 3 shows, covers nearly the same voiced frames as spectral-only placement. Frames redundant for the representation are thus safer than contributing ones but not the safest; local energy is what most effectively masks the watermark from the recognizer.

D. Ablations and Operational Boundary

Ablations (87 dB): fusion weights from $(0, 1)$ to $(0.75, 0.25)$ and adaptive versus fixed strength change SNR by at most 0.95 dB (86.05–87.00 dB); $R = 1$ and $R = 2$ gain 4.76 and

TABLE III
BIT ACCURACY (%) AFTER PROCESSING (200 UTTERANCES, 87 DB).

	AWGN (dB)				Scale	Resamp.	LPF	MP3	PCM
	40	30	20	10	0.9	8k	4k	128k	16-bit
Bit acc.	57.6	52.4	50.9	50.4	65.5	81.5	77.1	61.9	100.0

1.75 dB but fail exact recovery on 1 and 4 of 200 utterances, so $R = 3$ is kept.

The watermark survives storage as 16-bit PCM with 100% bit accuracy, so it persists in standard WAV files. Because each bit is the sign of a small displacement relative to the reference, additive noise, gain change, resampling, filtering, and compression lower bit accuracy to 50.4–81.5% (Table III), far from the exact match that verification requires; surviving such processing requires robust designs [22], [23]. SemBind therefore targets provenance verification of unmodified speech when the reference is available.

V. DISCUSSION

Implications. The results suggest three guidelines for content-aware placement. First, any content score should be gated by a local-energy term, as (6) shows that a fixed perturbation is most exposed where content is sparse; the multiplicative fusion of SemBind does this automatically. Second, placement should be compared at strengths where the recognizer responds (here ≤ 40 dB), since at transparent strength every rule is indistinguishable. Third, the fraction of changed transcripts is a simple, robust content measure for such comparisons, less affected than mean ΔWER by occasional recognizer hallucinations.

Limitations. The full-corpus placement sweep covers 2,620 utterances of read English speech from 40 speakers, but only one completed ASR model, Whisper tiny.en, is available. PESQ and STOI are ceiling-level in the clean-channel setting and therefore do not establish perceptual differences between placement rules. A completed second-ASR evaluation or a non-saturated perceptual measure is required before claiming model-independent content preservation. Extraction requires the reference, so SemBind addresses provenance verification of unmodified speech rather than detection after processing.

VI. CONCLUSION

Making the placement stage of a reference-based speech watermark explicit allows its effect on content to be measured in isolation. On the full 2,620-utterance sweep, spectral-only and joint placement substantially reduce transcript changes relative to high-contribution and random placement at stronger embedding levels. The frozen HuBERT contribution map is informative, separating safer from riskier frames significantly at 40–27 dB, but joint placement is statistically indistinguishable from spectral-only placement. Content-aware placement should therefore be energy-gated, while future work should test whether a revised semantic term can add value beyond local spectral suitability. Code and the selected coordinate lists will be released upon publication.

REFERENCES

[1] G. Hua, J. Huang, Y. Q. Shi, J. Goh, and V. L. L. Thing, "Twenty years of digital audio watermarking—a comprehensive review," *Signal Processing*, vol. 128, pp. 222–242, 2016.

[2] G. Hua, "Over-complete-dictionary-based improved spread spectrum watermarking security," *IEEE Signal Process. Lett.*, vol. 27, pp. 770–774, 2020.

[3] G. Hua, Y. Xiang, and L. Y. Zhang, "Informed histogram-based watermarking," *IEEE Signal Process. Lett.*, vol. 27, pp. 236–240, 2020.

[4] I. J. Cox, J. Kilian, F. T. Leighton, and T. Shamoon, "Secure spread spectrum watermarking for multimedia," *IEEE Trans. Image Process.*, vol. 6, no. 12, pp. 1673–1687, 1997.

[5] W. Bender, D. Gruhl, N. Morimoto, and A. Lu, "Techniques for data hiding," *IBM Syst. J.*, vol. 35, no. 3–4, pp. 313–336, 1996.

[6] B.-S. Ko, R. Nishimura, and Y. Suzuki, "Time-spread echo method for digital audio watermarking," *IEEE Trans. Multimedia*, vol. 7, no. 2, pp. 212–221, 2005.

[7] I. Natgunanathan, Y. Xiang, Y. Rong, W. Zhou, and S. Guo, "Robust patchwork-based embedding and decoding scheme for digital audio watermarking," *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 20, no. 8, pp. 2230–2238, 2012.

[8] K. Pavlović, S. Kovačević, I. Djurović, and A. Wojciechowski, "Robust speech watermarking by a jointly trained embedder and detector using a DNN," *Digit. Signal Process.*, vol. 122, p. 103381, 2022.

[9] G. Chen, Y. Wu, S. Liu, T. Liu, X. Du, and F. Wei, "WavMark: Watermarking for audio generation," *arXiv:2308.12770*, 2023.

[10] P. Li, X. Zhang, J. Xiao, and J. Wang, "IDEAW: Robust neural audio watermarking with invertible dual-embedding," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2024, pp. 4500–4511.

[11] M. K. Singh, N. Takahashi, W.-H. Liao, and Y. Mitsufuji, "SilentCipher: Deep audio watermarking," in *Proc. Interspeech*, 2024, pp. 2235–2239.

[12] R. San Roman, P. Fernandez, H. Elsahar, A. Défossez, T. Furon, and T. Tran, "Proactive detection of voice cloning with localized watermarking," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2024.

[13] H. Li, Z. Wu, X. Xie, J. Xie, Y. Xu, and H. Peng, "VoiceMark: Zero-shot voice cloning-resistant watermarking approach leveraging speaker-specific latents," in *Proc. Interspeech*, 2025, pp. 5108–5112.

[14] Y. He, H. Wang, Y. Qiu, and H. Cao, "ASSMark: Dual defense against speech synthesis attack via adversarial robust watermarking," *IEEE Signal Processing Letters*, vol. 32, pp. 1870–1874, 2025.

[15] Y. Feng, X. Zhang, F. Feng, G. Zhang, and L. Xu, "Robust and imperceptible watermarking framework for generative audio models," *IEEE Signal Processing Letters*, vol. 32, pp. 3196–3200, 2025.

[16] X. Fang, J. Li, J. Liu, and L. Zou, "Reinforced temporally sparse gating for high-fidelity speech watermarking," *IEEE Signal Processing Letters*, vol. 33, pp. 1936–1940, 2026.

[17] W.-N. Hsu, B. Bolte, Y.-H. H. Tsai, K. Lakhotia, R. Salakhutdinov, and A. Mohamed, "HuBERT: Self-supervised speech representation learning by masked prediction of hidden units," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 29, pp. 3451–3460, 2021.

[18] A. Baevski, H. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 12 449–12 460.

[19] National Institute of Standards and Technology, "Secure hash standard (SHS)," National Institute of Standards and Technology, Federal Information Processing Standards Publication FIPS PUB 180-4, 2015.

[20] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, "LibriSpeech: An ASR corpus based on public domain audio books," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process. (ICASSP)*, 2015, pp. 5206–5210.

[21] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2023, pp. 28 492–28 518.

[22] H. Liu, M. Guo, Z. Jiang, L. Wang, and N. Z. Gong, "AudioMarkBench: Benchmarking robustness of audio watermarking," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 37, 2024.

[23] Y. Özer, W. Choi, J. Serrà, M. K. Singh, W.-H. Liao, and Y. Mitsufuji, "A comprehensive real-world assessment of audio watermarking algorithms: Will they survive neural codecs?" in *Proc. Interspeech*, 2025, pp. 5113–5117.